Enhancing Detection of Left Ventricular Abnormalities on Chest Radiographs Using Modern Deep Learning Architectures

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I. Abstract

Severe left ventricular hypertrophy (SLVH) and dilated left ventricle (DLV) are structural heart problems that can lead to heart failure, arrhythmias, and higher mortality if they are not detected early. [11], [12], [13]. Echocardiography is the gold standard for measuring these abnormalities, but it is not always available [16]. Because chest X-rays (CXRs) are routine, cheap, and widely accessible, this project explores whether deep learning can detect SLVH and DLV directly from CXRs, using labels derived from echocardiography.

We used the CheXchoNet dataset [18], which contains 71,589 CXRs from 24,689 patients, paired with echocardiography measurements (IVSd, LVPWd, LVIDd) and patient metadata (age, sex, and other additional pacient information). The data comes from Columbia University Irving Medical Center and it was being collected between January 2013 and August 2018. The data and is imbalanced, most cases being healthy, so we built balanced subsets for better learning and evaluation.

First, we established a supervised learning baseline using the CheXchoNet dataset [18], which combines CXRs with echocardiography-derived labels and patient metadata such as age and sex. Following the methodology of Bhave et al. [2], we trained a DenseNet121-based model [21] to predict echocardiographic measurements and derive disease probabilities for SLVH and DLV. As stated earlier, a balanced subset of the dataset was constructed, and demographic information was incorporated into the model. This baseline demonstrated that chest radiographs contain meaningful information related to left ventricular abnormalities.

However, relying on explicit labels and demographic variables introduces several limitations. Echocardiography labels may be noisy, incomplete, or unavailable, while demographic attributes such as age and sex can be potentially masking the true visual signals present in CXRs.

To overcome these challenges, we explored self-supervised learning, which enables representation learning without explicit labels. We proposed using a Vision Transformer (ViT) [25] trained in a self-supervised manner on CXRs only, without access to age, sex, or heart measurement labels during representation learning. Vision Transformers [25] are particularly well suited for this task due to their ability to model long-range dependencies and capture global image structure. The Vision Transformer [25] (ViT-Small) [32] was used as a frozen feature extractor, initialized from self-supervised Medical-MAE pretraining [33] (trained on approximately 300.000 chest radiographs). On top of the frozen encoder, we trained a small MLP classifier for binary prediction, SLVH/DLV and healthy, depending on the task at hand. We tested preprocessing and augmentation choices separately each time to see which techniques achieve the best results and the final pipeline used per-channel normalization with mean 0.5 and std 0.5, and some augmentations such as random crop and resize, horizontal flip, small rotations. These were the ones that gave the best outcome.

The DenseNet121 [21] baseline obtained test AUROC values of about 0.73 for SLVH, 0.75 for DLV, and 0.79 for the composite outcome, which suggests that standard CXRs contain useful information for identifying these conditions when the labels are derived indirectly from echocardiography measurements. But the most important thing was to see if the CXRs alone were enough to diagnose the left ventricular abnormalities studied. The transformer method improved performance on both tasks, reaching AUROC 0.7807 for SLVH and AUROC 0.7887 for DLV. These improvements were also influenced by the training pipeline: preprocessing and augmentation choices mattered and improved slightly the outcome, with simple, consistent normalization producing more stable learning than alternatives.

This project shows that deep learning can detect SLVH and DLV from routine CXRs, and that modern transformer features learned through self-supervision can outperform a strong supervised DenseNet121 [21] baseline, which consists in a great start for future research. By freezing a Medical-MAE [33] pretrained ViT encoder and training a small classifier, we tried to reduce dependence on noisy echocardiography-derived regression targets and to avoid explicitly using age

or sex during representation learning, while still improving AUROC for both SLVH and DLV.

II. Introduction

Deep learning enables computational models to understand and infer on complex data, through a series of processing layers, each specialized on learning increasingly abstract features. These network architectures have been able to consistently outperform traditional machine learning algorithms across various domains, including speech recognition, visual object recognition, object detection, and medical analysis. By using the backpropagation algorithm to correct internal parameters, a deep learning network is able to identify intricate patterns and hierarchical representations in large data sets [5].

Narrowing down to the medical imaging field, various machine learning models have demonstrated accurate diagnosis potential across different tasks and data formats. In radiology, CNNs [20] have been used to detect pneumonia, lung cancer, and cardiomegaly on x-rays and computer tomography (CT) scans [6], [7]. Similar architectures have enabled automated recognition of diabetic retinopathy in retinal fundus images [8] and skin cancer in dermoscopic photographs [9]. Within the field of cardiology, recent work has shown that deep learning methods can effectively detect reduced ejection fraction, valvular heart disease and LV hypertrophy from cardiac tests such as echocardiograms (TTEs), electrocardiograms (EKGs) and chest radiographs (CXRs) [1], [2].

Severe Left Ventricle Hypertrophy (SLVH) and Dilated Left Ventricle (DLV) are common heart structural abnormalities that are manifestations of cardiac remodeling. Both conditions reflect the heart's adaptive response to hypertension, valvular disease, ischemic injury, or genetic predisposition. These changes progressively impair ventricular function, leading to heart failure, arrhythmias, and increased mortality if left unrecognized [11], [12], [13]. These heart conditions can be fully reversible if they are detected in their initial stages, making timely diagnosis potentially life-saving [10]. Early treatment of asymptomatic patients with left ventricular dilation was shown to significantly reduce progression to symptomatic heart failure and mortality.

The first iteration of applying CNNs for diagnosing ventricular abnormalities focused on identifying the target region of the left ventricular posterior wall [17], on which image processing tehniques were applied to obtain a result [1]. This study represented a major breaktrough in medical imaging, demonstrating the potential of machine learning models for cardiac assessment. However, it soon became clear that several aspects could be improved, starting from the architecture used to the dataset itself. TTEs come with a trade-off: while being among the most clinically informative, they are not only costly and time-consuming to acquire, but also computationally intensive to process and train on, motivating the search for alternative data collections [2], [15].

Echocardiograms are the most commonly used tests to examine the heart because they give clear, detailed images of its structure and function [16]. However, chest X-rays are faster, cheaper, and easier to access since they do not need special equipment and can still show important signs of heart problems [2], [14], [15].

Recent research used this information and introduced the idea of using artificial intelligence to analyze CXRs and detect the same structural heart problems [2], [3], [4]. Building on this approach, a deep learning model has been developed, capable of identifying SLVH and DLV directly from routine CXRs, helping diagnose people before ever getting an echocardiogram done [2].

Expanding on the above mentioned paper, it became evident that the initial model was susceptible to noise and dataset shifts, as it had been trained and evaluated on a carefully curated collection of radiographs. To address this limitation and ensure accurate results on inputs with a higher variance, a follow-up study published in November 2024 introduced synthetic data created by generative artificial intelligence and diffusion models. The initial dataset has been expanded with this artificial data, the model being trained and tested on the original collection of x-rays, the synthetic one, and a mixture of both, the latter achieving the highest accuracy, demonstrating the potential of generative AI for improving deep learning performance in medical imaging tasks [3].

The Bhave et al. (2024) study [2] used both CXRs and patient demographics (age, sex) to detect SLVH and DLV. These variables were included to improve diagnostic accuracy and align the model with clinical patterns found in TTEs. In contrast, a later paper written in 2025 [4] proposed a simpler approach that classified SLVH directly from chest X-rays alone, without using demographic data. Their model focused on the image features, showing that high accuracy could still be achieved while avoiding possible bias from patient information. Later, they showed that even when trained only with binary labels, it still internalized important physiological and demographic patterns related to SLVH.

III. Related work

Each study mentioned used different methods to detect cardiac abnormalities, reflecting how the technology has evolved over time. These are the methods in chronological order:

Jian et al. (2020) [1] combined a CNN [20] with 7 layers with image processing to measure the left ventricular posterior wall thickness (LVPWT) from echocardiogram images. The CNN included convolutional, pooling, and fully connected layers with ReLU activation and dropout regularization to prevent overfitting. It's role was to locate the posterior wall region. Following this, Non-Local Means filtering and morphological operations were used to reduce noise while preserving important edges. They then applied a custom edge-detection algorithm to calculate the wall thickness. In this study, a dataset composed of around 2000 echocardiograms images from 50 different patients was used. The measurement system based on CNNs was evaluated using mean relative error and accuracy against manual clinical measurements, achieving percentages below 15%, but the dataset was very small, so it was hard to generalize. The method also relied on manual

processing and was not fully automated. There is still room for improvement, as this study is limited by low dataset diversity and the absence of advanced transformer models capable of processing 3D TTE data.

Bhave et al. (2024) [2] introduced the backbone of SLVH and DLV detection using deep learning. This study was based on the CheXchoNet dataset [18], which combines 71,589 CXRs and corresponding TTEs from 24,689 patients at Columbia University. Echocardiographic measurements (IVSd, LVPWd, LVIDd) were used to label cases of SLVH and DLV. The dataset was split into training, validation, and test sets, with an additional external validation dataset from Stanford University to avoid biases. The chosen model was based on a DenseNet121 convolutional architecture [21], because of it's image detail extraxtion properties. The images were cropped, resized and then enhanced with the histogram equalization [19] technique before being combined with age and sex as inputs. Also, it used LayerCAM saliency maps [22] to show which parts of an image influenced predictions. Performance was measured using AUROC, AUPRC, sensitivity, and specificity, and the model outperformed 15 radiologists in detection accuracy. Despite strong results, limitations included dependence on binary labels taken from echocardiograms and potential demographic biases, as age and sex were used as model inputs.

Streiffer et al. (2024) [3] extended this work by implementing a diffusion probabilistic model (DDPM) [23] trained on the same CheXchoNet dataset [18] that worked by adding Gaussian noise to CXRs and then learning to reverse the process while conditioning on age, sex, IVSd, LVIDd and LVPWd. The trained diffusion model generated a synthetic dataset with a higher proportion of positive cases and younger patients, compensating for the few cases in the original dataset. They also used models based on a DenseNet121 CNN architecture [21] that were trained on the real, synthetic, and combined datasets. Models trained on the combined data achieved the best results, improving accuracy scores, particularly for younger patients. Evaluation focused on AUROC and AUPRC for models trained on the real, synthetic, and combined datasets. A downside of this study consists of the synthetic nature of generated data, which may not perfectly represent true CXRs, needing clinical validation before being used to train and test the model.

Pal et al. (2025) [4] proposed the most recent advancement, buliding a model to detect exclusively SLVH, from CXRs, without using demographic or anatomical data. They used ResNet18 [24] and a Vision Transformer (ViT) [25] pretrained with masked autoencoding (MAE) [26]. After freezing the pretrained encoder, a classifier head was trained on the labeled CheXchoNet dataset [18], same dataset as the previous 2 studies. The model was tested with Mutual Information Neural Estimation (MINE) [27], and the result was that it could still learn patterns such as age, sex, and heart structure features from only the images. Performance was assessed using AUROC, accuracy, F1-score, and calibration metrics across internal and external test sets. The study also evaluated the model with MINE and saliency maps. Some limitations are the usage of a smaller dataset size after class balancing, use of only SLVH labels (no DLV), and the fact that the model's pretraining was done on external datasets, which may not fully match the clinical data distribution.

IV. Methodologies

For the baseline, we built a supervised model using the CheXchoNet dataset [18]. Since the original dataset is highly imbalanced, we extracted a balanced subset with 5,000 healthy cases and an equal number of SLVH and DLV cases, because the DLV cases were scarcer, plus all available combined-pathology samples to ensure representation across all classes. Images were already standardized in the dataset pipeline, resized to 224x224 and had Gaussian Noise added, and we additionally applied dataset-level normalization using the overall mean and variance to improve generalization. Age was standardized based on training-set statistics and sex was one-hot encoded. We split the data using a strict patient-level split to prevent leakage, allocating 90% for training and 10% for testing, with 5% of the total used as a validation set.

For the model, we used DenseNet121 [21] pretrained on ImageNet as the backbone, and replaced its final classification layer with a feed-forward head that outputs mean and log-variance for three independent Gaussian distributions, one per echocardiography measurement (IVSd, LVPWd, LVIDd). DenseNet [21] produces a 1024-dimensional global average pooled feature vector, which we concatenated with the processed age and sex features, and then passed through a final prediction layer. Training followed the reference approach by supervising the predicted means and variances with a Gaussian negative log-likelihood loss [31], optimized using Adam [34] for stability. After training, the regressed Gaussian outputs were converted into disease probabilities by computing the probability that each measurement exceeds clinically defined thresholds (IVSd > 1.5 cm, LVPWd > 1.5 cm, LVIDd > 5.9 cm), these conditions were taken from the study we tried to recreate, Bhave et al. (2024) [2]. The SLVH probability was computed by combining the IVSd and LVPWd probabilities with the formula for union of independent events, while the DLV one was derived directly from LVIDd and a composite probability was obtained by combining SLVH and DLV in the same way.

For our own contribution, we switched to a transformer-based setup that learns image features without using demographic variables. For each task, SLVH and DLV, we built two balanced binary datasets with the same number of healthy and disease cases, one for SLVH and one for DLV, and we removed samples with combined pathologies to see if the diseases can be recognized separately. We used stratified splits into 90% training, 5% validation, and 5% test. All images were normalized per-channel using a fixed mean of 0.5 and standard deviation of 0.5, and this choice gave the most stable improvements overall. During training we applied data augmentation to reduce overfitting and simulate realistic acquisition changes: random resized crop (scale 0.8–1, ratio 0.9–1.1), horizontal

flip (p = 0.5), and small rotations (15 degrees). Validation and test images used the same normalization but no augmentation.

For the model, we used ViT-Small [32] as a fixed feature extractor with 12 transformer layers and embedding size 384, initialized with Medical-MAE [33] self-supervised pretrained weights trained on about 300.000 CXRs. We removed the original head and froze all transformer encoder weights to keep the pretrained representation and limit overfitting on the smaller balanced subset. Without the freezing of the weights, the overfitting started around epoch 3, meaning it was learning way too quickly and not generalizing. On top of the frozen encoder, we trained a small MLP classifier with fully connected layers 384, to 512, to 128, to 1, using ReLU after hidden layers and dropout 0.2 for regularization. Training used binary cross-entropy with logits, optimized with AdamW [34] with a learning rate of 1e-4 and weight decay. We added a warm-up of 2 epochs followed by cosine learning-rate decay, trained up to 10 epochs while selecting preprocessing and augmentation techniques, and then trained 20 epochs for final results. We monitored validation AUROC each epoch, saved the best model, and used early stopping when validation performance stopped improving.

V. Dataset description

The CheXchoNet dataset [18] consists of 71,589 unique chest X-rays from 24,689 different patients alongside key echocardiography measurements indicative of left ventricular hypertrophy and dilated left ventricle (ivsd, which measures the septal wall thickness; lvpwd, the thickness of the left walls of the heart; and lvidd, the overall size of the left ventricle). The dataset also includes temporal information for each radiograph (cxr_time_offset, cxr_year), patient demographic attributes (age, sex), and additional metadata (heart_transplant, lung_transplant, pacemaker_or_icd). This comprehensive multimodal resource enables the development and evaluation of advanced models for detecting left ventricular abnormalities.

All samples have been collected from the Columbia University Irving Medical Center, including patients who had both an CXR and echocardiogram, providing clinically confirmed cardiac diagnoses, from January 2013 to August 2018.

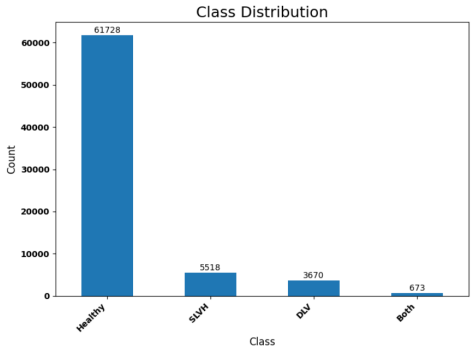

Fig. 1. Class distribution across patient groups.

The dataset [18] exhibits an uneven class distribution, containing 86.2% Healthy samples, 7.7% SLVH, 5.1% DLV, and 0.9% patients with both diseases, a key aspect to consider when developing predictive models.

Demographically, the dataset [18] covers a broad range of age groups, with the highest concentration occurring between 50 and 80 years. The gender classes are balanced, eliminating the risk of gender-related bias in subsequent model evaluation.

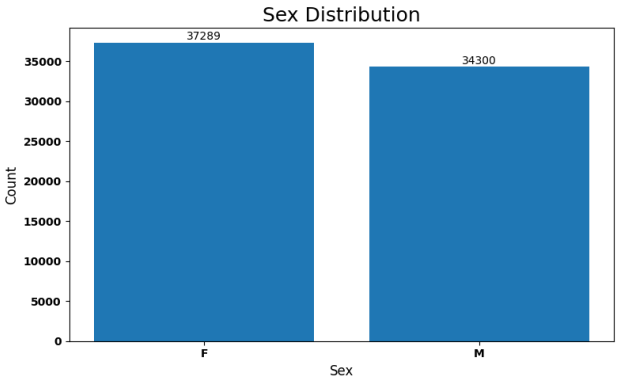

Fig. 2. Gender distribution across all patients.

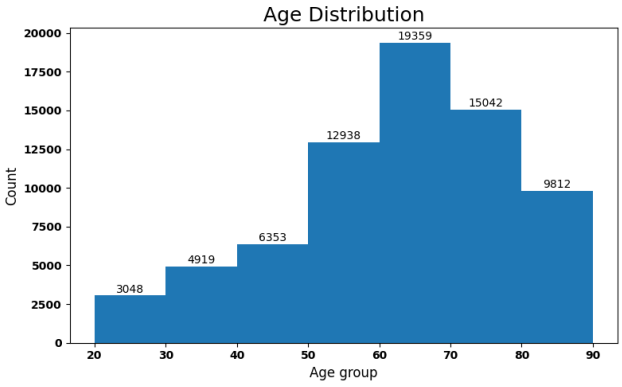

Fig. 3. Age distribution across all patients.

VI. Data preprocessing for the baseline

Several standard preprocessing procedures steps have already been applied to the CheXchoNet data collection [18], in order to make the data tractable and easy to use for downstream application, such as downsampling to 224-by-224 pixel images using bicubic interpolation and contrast-limited adaptive histogram equalization.

In addition, the demographic variables were preprocessed by standardizing age using the mean and standard deviation computed from the training set and by one-hot encoding the sex attribute. To create a reliable baseline backed up by scientific research, each step of the Bhave et al. paper [2] has been followed, including the normalization of the dataset with the overall mean and variance and the addition of gaussian noise using normal distribution with mean 0 and variance 0.05, both aiming to improve generalization performance.

For the purposes of this study, a balanced subset consisting of 5,000 healthy cases and an equal number of SLVH and

DLV samples, alongside all the available combined-pathology subjects has been extracted, aiming at fair and unbiased model training and evaluation across all diagnostic categories.

Following the construction of the balanced subset, the data were partitioned into training, validation, and test sets using a patient-level split to prevent information leakage across partitions. Specifically, 90% of the samples were allocated to the training set and 10% to the test set, out of which 5% form the validation set, ensuring that no images from the same patient appear in more than one subset.

VII. Baseline model

The architecture applied uses DenseNet121 [21] as the backbone, having the last layer responsible of classifying the input replaced with a feed forward layer that generates mean and variance parameters for three independent univariate normal distributions, one for each echocardiographic measurements. Subsequently, these distributions are converted to probabilities for each of the three binary labels indicating structural abnormalities.

The DenseNet model [21], which consists of 4 dense blocks and 3 transition layers, is pretrained on ImageNet, and produces a 1024-dimensional representation via global average pooling; this image feature vector is then concatenated with the standardized age and one-hot encoded sex inputs, and passed through a fully connected prediction layer. Model optimization follows the approach described by Bhave et al. [2], using the Adam optimizer [34] for stable gradient-based learning and a Gaussian negative log-likelihood [29] loss to supervise the estimation of the mean and variance parameters for each echocardiographic measurement.

To convert the model's regression outputs into disease probabilities, the same probabilistic formulas used in the base study are applied. For each echocardiographic measurement, IVSd, LVPWd, and LVIDd, the network predicts a mean value and a log-variance, defining a Gaussian distribution that reflects both the estimated value and its uncertainty. Using these distributions, we compute the probability that each measurement exceeds the clinically established thresholds: IVSd > 1.5 cm, LVPWd > 1.5 cm, and LVIDd > 5.9 cm, associated with left ventricular hypertrophy or dilation. The probability of SLVH is obtained by combining the IVSd and LVPWd probabilities using the standard formula for the union of independent events, while DLV is derived directly from the probability that LVIDd exceeds its threshold. A composite probability is then formed by combining SLVH and DLV in the same manner.

VIII. Evaluation metrics for the baseline

The model shows a clear upward trend in AUC for all three tasks, SLVH, DLV, and Composite, indicating that it successfully learns meaningful features from chest radiographs. The Composite prediction performs best overall, while SLVH and DLV show more modest improvements. Training loss decreases steadily, while validation loss remains more variable, likely due to label noise in echocardiographic measurements and the limited size of the reduced subset used for training. These patterns are reflected in the final test results, where the model reaches an AUC of approximately 0.73 for SLVH, approximately 0.75 for DLV, and approximately 0.79 for the composite outcome, with a test loss of approximately 0.41. Overall, the model generalizes reasonably well given the limited dataset and the complexity of the targets. Importantly, these values are close to the performance reported in the original article, which achieved composite AUCs around 0.80–0.83 on the full dataset. Considering that our model is trained on a smaller subset, the results obtained indicate behavior consistent with the findings of the original study.

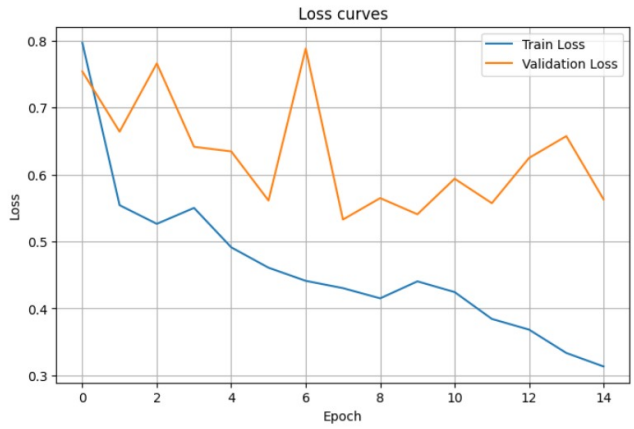

Fig. 4. Loss vs Epoch.

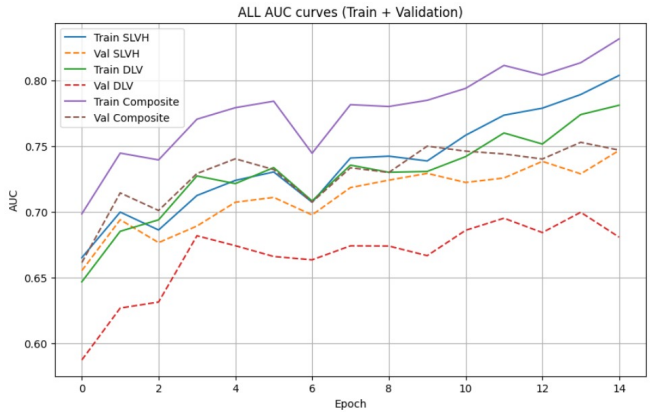

Fig. 5. AUC vs Epoch.

After adjusting the decision thresholds to match a target sensitivity of 66%-70%, as done in the original study, the model effectively identifies positive cases across all tasks. SLVH and DLV show moderate but consistent performance, while the Composite results are the strongest. Composite performs best because it integrates information from both structural abnormalities, making it a more reliable target. Precision is lower for SLVH and DLV, but the higher recall values indicate that the model successfully detects a large proportion of true positive cases. Overall, the classification scores reflect a balanced behavior, with the calibrated thresholds enabling

the model to maintain strong sensitivity while preserving reasonable precision.

TABLE I
PRECISION, RECALL, AND F1-SCORES FOR EACH PREDICTION TASK.

Task	Precision	Recall	F1-score
SLVH	0.420	0.698	0.524
DLV	0.446	0.712	0.548
Composite	0.718	0.701	0.710

IX. DATA PREPROCESSING FOR SELF-SUPERVISED LEARNING

We followed the same preprocessing pipeline for both DLV and SLVH experiments. For each task, we created a balanced dataset by selecting an equal number of healthy cases and disease-specific cases (DLV-only or SLVH-only). Samples with combined pathologies were excluded.

Each dataset was split into training, validation, and test sets using stratified sampling to keep the class distribution consistent. The training set contains 90% of the data, while the remaining 10% is equally divided between validation and test sets. All images were normalized on a per-channel basis using a fixed mean of 0.5 and standard deviation of 0.5. The normalization step improved final AUROC test scores for both SLVH and DLV significantly.

During training, data augmentation was applied to increase data variability and reduce overfitting. This included random horizontal flipping, small random rotations, and random cropping and resizing, which simulate realistic differences in patient positioning and image acquisition. The validation and test sets were processed using the same normalization steps but without any augmentation, in order to provide a consistent and unbiased evaluation of model performance.

The following tables report the performance of the model under different preprocessing strategies, where each experiment uses only one preprocessing method at a time.

TABLE II
SLVH CLASSIFICATION UNDER DIFFERENT PREPROCESSING

Metric	None	Normalize	Per-image std	Gaussian Blur
AUROC	0.7622	0.7694	0.7148	0.7190
Precision	0.6859	0.6858	0.6715	0.5744
Recall	0.6884	0.7355	0.5036	0.9094
F1-score	0.6872	0.7098	0.5756	0.7041

TABLE III
DLV CLASSIFICATION UNDER DIFFERENT PREPROCESSING

Metric	None	Normalize	Per-image std	Gaussian Blur
AUROC	0.7495	0.7637	0.7787	0.7513
Precision	0.6864	0.7016	0.7069	0.6901
Recall	0.6304	0.7283	0.6685	0.6413
F1-score	0.6572	0.7147	0.6872	0.6648

For SLVH, standard normalization consistently improves performance across all metrics compared to the baseline and was chosen as the best preprocessing method for the remaining experiments. Per-image standard deviation normalization leads to lower AUROC and F1-score, while Gaussian blur increases recall but at the cost of very low precision, both indicating overfitting.

Per-image standard deviation normalization achieves the highest AUROC, but results in lower precision and F1-score, suggesting less balanced predictions. Gaussian blur does not improve performance, likely due to the loss of fine structural details. Simple normalization provides the most stable performance across all metrics and was selected as the main preprocessing method for the DLV.

The following tables report the performance of the model under different augmentation strategies, where each experiment uses only one augmentation method at a time.

TABLE IV
SLVH CLASSIFICATION UNDER DIFFERENT AUGMENTATION TECHNIQUES

Metric	Crop	VFlip	HFlip	Rotation	ColorJitter
AUROC	0.7744	0.7695	0.7713	0.7848	0.6658
Precision	0.6920	0.6820	0.6887	0.7137	0.7123
Recall	0.7246	0.7536	0.7536	0.6775	0.3768
F1-score	0.7080	0.7160	0.7197	0.6952	0.4929

TABLE V
DLV CLASSIFICATION UNDER DIFFERENT AUGMENTATION TECHNIQUES

Metric	Crop	VFlip	HFlip	Rotation	ColorJitter
AUROC	0.7507	0.7398	0.7599	0.7519	0.5951
Precision	0.6856	0.6567	0.7011	0.7063	0.8824
Recall	0.7228	0.7174	0.7011	0.6141	0.0815
F1-score	0.7037	0.6857	0.7011	0.6570	0.1493

For both SLVH and DLV, random resize cropping, horizontal flipping and random rotation have the best overall performance. Vertical flipping and color jitter do not provide improvements, likely due to anatomical inconsistency and excessive intensity variation. Based on these observations, random resized cropping with scale from 0.8 to 1 and ratio from 0.9 to 1.1, horizontal flipping with probability of 50%, and random rotation with 15 degrees were selected as the final augmentation strategy for both SLVH and DLV experiments.

X. OWN CONTRIBUTION

In this work, we employ a ViT-Small architecture [32] as a fixed feature extractor. The transformer encoder, consisting of 12 layers with an embedding dimension of 384, was initialized using self-supervised Medical-MAE pretrained weights [33], obtained from pretraining on approximately 0.3M chest radiographs. The original classification head was removed, and all encoder parameters were frozen during training to preserve learned representations and mitigate overfitting.

On top of the frozen encoder, we added a multi-layer perceptron (MLP) classifier to perform binary classification. The classifier consists of three fully connected layers with decreasing dimensionality (384, 512, 128, then 1). ReLU activation functions are used after each hidden layer, and dropout with a rate of 0.2 is applied to improve regularization.

The final layer outputs a single logit corresponding to the probability of disease presence.

The model was trained using the binary cross-entropy loss with logits, which is suitable for binary classification tasks. Optimization was performed using the AdamW [34] architecture with a learning rate of 0.0001 and weight decay for regularization. Training was carried out for a maximum of 10 epochs when trying to find the best preprocessing and augmentation strategies, then for the final test results, a number of 20 epochs was used. To stabilize optimization in the early stages, a learning rate warm-up was applied during the first two epochs, followed by a cosine decay schedule for the remaining epochs. Model performance was evaluated after each epoch using loss and AUROC on the validation set. The model with the highest validation AUROC was saved, and early stopping was used to terminate training if no improvement was observed for several consecutive epochs.

XI. Results of own contribution

The experimental results demonstrate that the proposed approach outperforms the baseline in the diagnosis of both SLVH and DLV. The improved performance on SLVH, a condition previously investigated in related studies, serves as a validation of the model's reliability, while the gains observed for DLV highlight the potential of the proposed method for automated medical image analysis.

TABLE VI
Precision, Recall, and F1-scores for the test subset

Disease	AUROC	Precision	Recall	F1-score
SLVH	0.7807	0.6990	0.7572	0.7270
DLV	0.7887	0.7074	0.7074	0.7170

For SLVH, the transformer architecture obtains an AUROC of 0.78, which is comparable to the approximately 0.80 reported in prior [4], indicating consistent performance with established results. This performance exceeds the AUROC of 0.73 obtained using the DenseNet121 baseline [21], supporting the effectiveness of the proposed approach.

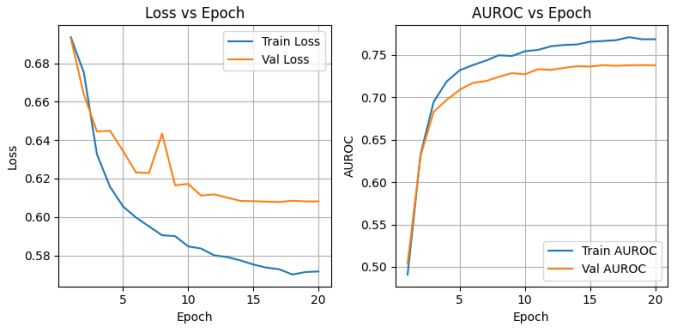

Fig. 6. SLVH train and validation results

For DLV, the above mentioned model achieves an AUROC of approximately 0.79, outperforming the baseline, which attains an AUROC of 0.75. Unlike SLVH, DLV has not been extensively investigated in prior studies, and these results suggest that vision transformers [25] are effective in identifying pathologies that can be inferred from chest X-ray images even in the absence of task-specific pretraining.

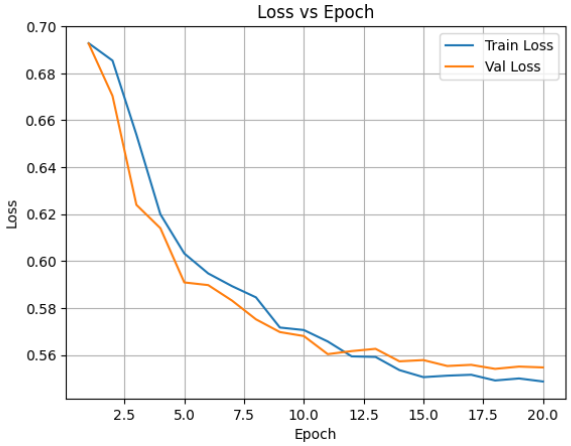

Fig. 7. DLV train and validation loss results

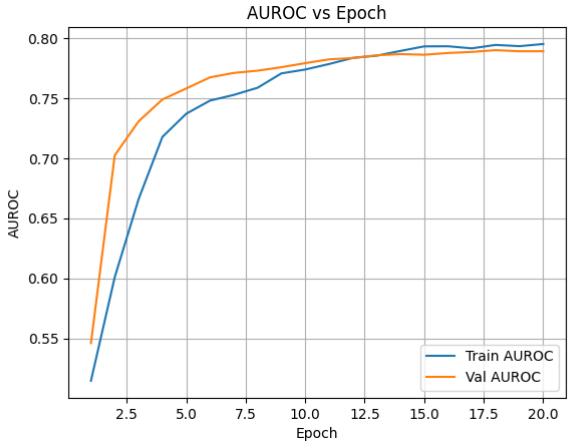

Fig. 8. DLV train and validation AUROC results

In addition, the application of data augmentation during training proved beneficial, leading to AUROC improvements of approximately 0.01-0.03 across both tasks. These gains are notable given the limited size of the training subset, highlighting the importance of augmentation strategies in small-sample medical imaging scenarios.

XII. Limitations

At the same time, the limitations are real and they define how far we can trust these resulted numbers. The balanced subset is helpful for fair training but it does not reflect real world situations, so the same model could behave differently when healthy cases dominate and when decision thresholds must match clinical workflow. The labels come from echocardiography measurements and clinical cutoffs, so noise, timing mismatch between exams, and borderline measurements can

blur the ground truth and make validation behaviour unstable, which can alter the performance even if the model is improved. A limitation of this whole field is that we are trying to detect subtle heart structure changes using a test that was not designed to measure the heart directly, so the signal in CXRs is real but weak and indirect, which means small shifts in data quality, acquisition protocol, or patient mix can influence performance in ways that are hard to predict.

Generalization is another concern because the dataset is collected from one center and includes preprocessing choices like resizing and contrast enhancement, so performance may shift across hospitals, devices, and acquisition styles unless external testing is done. The transformer training strategy also has a tradeoff, since freezing the encoder protects against overfitting but may prevent the model from adapting to the most disease specific cues, especially for dilation where visual signals can be subtle and easily mixed with other elements. Finally, avoiding explicit demographics does not automatically remove bias, since prior work shows image only models can still encode age and sex related information from the radiograph itself, so interpretability must be measured.

Class imbalance is also a problem in real life situations because healthy cases dominate, so balanced training is useful for learning but it can give an optimistic picture of how thresholds will behave in real screening workflows. Bias and shortcut learning are another ongoing concern. Adding age and sex can improve accuracy but also increases the chance that the model relies on demographic patterns instead of true visual cues, and even image only models can still encode demographic and anatomical components. A previous study also shows that attempts to fix the discrepancies in real world data using synthetic data can help, but they introduce their own limitation because generated images may not perfectly match real clinical radiographs and need careful validation before they can be trusted as training evidence. At the same time, our results also highlight remaining gaps. Performance is still bounded by label uncertainty, by single center data, and by the tradeoff of freezing a pretrained encoder that may not fully match the target distribution, so the next logical step is external validation and careful fine tuning that adapts the representation without reintroducing overfitting or demographic shortcuts.

XIII. Conclusions and future improvements

Overall, this project supports the idea that routine CXRs contain real and usable information about left ventricular structural diseases, and it shows that better feature learning can matter just as much as the final classifier you put on top. With the supervised DenseNet121 [21] baseline that first predicts echocardiography measurements and then turns them into probabilities for SLVH and DLV, we saw a decent performance on a reduced, balanced subset, around 0.73 AUROC for SLVH, around 0.75 for DLV, and around 0.79 for the composite label, which is important because it suggests the pipeline is learning meaningful image patterns rather than fitting random noise, even though the original clinical scale setting is larger and typically reports stronger composite scores.

The transformer approach then pushed performance further by changing what is learned and how it is learned, using a Vision Transformer [25] encoder pretrained with MAE [33] on a large CXRs collection and keeping that encoder frozen while training a small MLP head, which is a more data efficient way to reuse general radiographic structure and reduces the risk of overfitting when the labeled subset is limited. Under this setup, results improved for both conditions to about 0.7807 AUROC for SLVH and about 0.7887 for DLV, and simple augmentations delivered small but meaningful extra gains to the final predictions. The comparison between approaches is also really useful because the DenseNet121 [21] baseline explicitly combines image features with age and sex and is trained directly against echo derived targets, so it can benefit from strong supervision but it may also partly rely on demographic shortcuts, while the self supervised transformer shows that you can still reach higher accuracy by learning stronger image representations without using demographics during representation learning.

From the task-level comparison, we found that SLVH and DLV do not behave identically, even when they are learned from the same image dataset and using the same procedures. In our experiments, DLV was slightly easier to separate than SLVH across both models. This gap is plausible clinically because dilation can introduce more global changes that influence the overall cardiac silhouette and thoracic proportions, while hypertrophy can be more subtle on a frontal radiograph and may depend on secondary signs that are weaker and more easily confounded. At the same time, the improvement brought by the transformer was not disease-specific, as it lifted both labels by a similar margin compared with DenseNet121 [21], which suggests the representation is capturing general chest structure cues that help across multiple forms of LV remodeling rather than overfitting to one in particular.

Beyond AUROC, the models also highlight practical differences between the two conditions. The precision and recall balance indicates that the models can reach reasonable sensitivity on both tasks, but there is still a tradeoff in how confidently positives are identified, which is especially relevant if the goal is screening. In other words, the results support the idea of using CXRs for possible early detection, where the model flags higher-risk studies for follow-up echocardiography, but they also show that the models are not yet at a level where they can replace echo assessment, especially when higher precision is needed for identification.

A next step in our research could be trying to use more real world scenarios, by training and evaluating on the full imbalanced dataset, adding strong external validation across sites, exploring careful fine tuning that adapts the pretrained transformer without overfitting, and expanding beyond single label prediction toward multitask learning with so the model becomes not only accurate but also reliable enough to support clinical decisions. Also, data must be aquired from multiple hospitals and using different techniques of obtaining CXRs, to improve generalization and test the model's integrity.

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